

A NEW SIMILARITY-BASED RELATIONAL KNOWLEDGE DISTILLATION METHOD

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ABSTRACT

The previous relation-based knowledge distillation methods tend to construct global similarity relationship matrix in a mini-batch while ignoring the knowledge of neighbourhood relationship. In this paper, we propose a new similarity-based relational knowledge distillation method that transfers neighbourhood relationship knowledge by selecting K -nearest neighbours for each sample. Our method consists of two components: Neighbourhood Feature Relationship Distillation and Neighbourhood Logits Relationship Distillation. We perform extensive experiments on CIFAR100 and Tiny ImageNet classification datasets and show that our method outperforms the state-of-the-art knowledge distillation methods. Our code is available at: <https://github.com/xinxiaoxiaomeng/NRKD.git>.

Index Terms— Knowledge Distillation, Similarity Relationship Knowledge

1. INTRODUCTION

As a simple and efficient model compression approach, knowledge distillation (KD) [1] has been widely used in different tasks, including visual recognition tasks such as image classification [2], and natural language processing tasks such as machine translation [3]. Up to now, the knowledge used for distillation can be divided into three categories: 1) response-based knowledge distillation mimics the logits of the teacher network [2, 4]; 2) feature-based knowledge distillation tries to enable the student to learn the intermediate feature representations of the teacher network [5, 6]; and 3) relation-based methods facilitate the network to learn the similarity relationship between different samples [7, 8, 9, 10].

The previous relation-based distillation methods [7, 8, 9, 10] have two main limitations: 1) Exploring complex similarity relationship. A light student network has difficulty in maintaining the same global relationship as a cumbersome teacher network due to the capacity mismatch. [11] has proved that the performance of the student network declines when the capacity gap between the teacher and student

networks is too large. For a student network, it is easy to distinguish samples of the class label “tree” from samples of the class label “dog”, and it might be difficult to distinguish between “dog” and “cat”. The problem is that they have relatively similar feature representations, making it very difficult and also very important to explore the relationship between two samples; 2) Failing to capture rich feature representation information. Specifically, each element in the similarity matrix usually indicates the similarity of feature representations between two samples. However, there are more information in the feature representations between two different images, rather than a single value as the similarity information for distillation.

To address the above-mentioned limitations, we propose a new similarity-based relational knowledge distillation method to transfer the neighbourhood relationship knowledge. Specifically, we first construct the similarity matrix using the feature representations of the teacher network for neighbour selection (K -nearest neighbours for each sample are selected from a mini-batch); we then construct a neighbourhood relationship using the feature maps and logits of each sample and its K -nearest neighbours. Instead of directly distilling the similarity matrix, we construct the rich similarity information from the output of the convolutional blocks and the classifier. The similarity between samples is not mapped to a specific value. We evaluate the proposed method by conducting extensive experiments on CIFAR100 [12] and Tiny ImageNet [13] classification datasets. Our method outperforms recent SOTA knowledge distillation methods.

2. METHOD

The framework of our method is shown in Fig. 1. We construct neighbourhood rather than global relationship by selecting K -nearest neighbour samples of each sample. We build rich similarity information between samples for distillation, not only a similarity value. Our method transfers two parts of knowledge, the neighbourhood feature relationship and the neighbourhood logits relationship.

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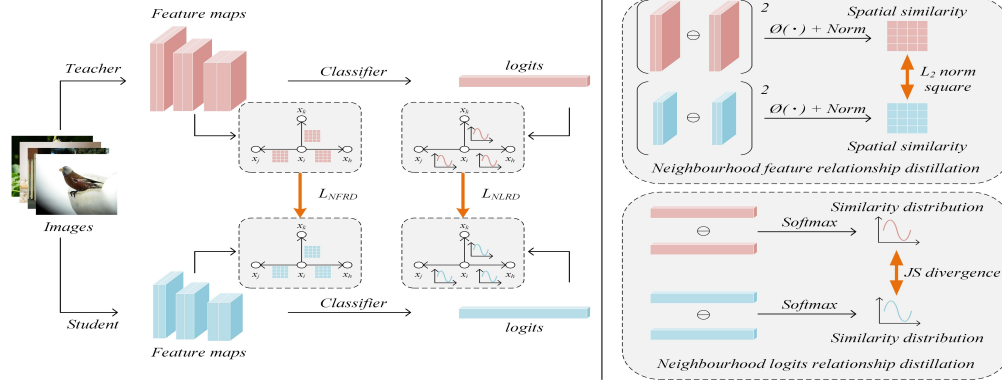


Fig. 1. The overview of the proposed method. Samples x_j, x_k, x_h are three selected neighbours of sample x_i . $\ominus$ represents element-wise minus. $\phi(\cdot)$ represents the channel-wise average pooling. Our method constructs both spatial similarity information from intermediate feature maps and similarity distribution information from the final output logits as the neighbourhood relationship knowledge.

2.1. Preliminary

Knowledge distillation was proposed by Hinton [2]. We define the training data as $D = \{x_i, y_i\}_{i=1}^N$, where N is the number of samples, and y_i is the true label of the sample x_i , the subscript i indicates the index of each sample. We define the output logits of the sample x_i for the teacher and student networks as z_i^T and z_i^S , respectively. [2] introduced a temperature τ to obtain soft targets as additional supervision of the student network. The distillation loss is shown in Eq. (1):

$$L_{KL} = \frac{\tau^2}{N} \sum_{i=1}^N KL(\text{softmax}(z_i^T/\tau), \text{softmax}(z_i^S/\tau)). \quad (1)$$

Kullback-Leibler (KL) is used to measure the probability distribution differences between the teacher and student networks. According to Eq. (1), we can find that the student network learns the soft targets of the teacher network for each sample directly, assuming that the samples are independent of each other.

2.2. Neighbour Selection

Generally, the teacher network performs better than the student network. Therefore, we choose the neighbours of each sample according to the output logits of the teacher network. We construct the similarity matrix D , which we use to select the neighbours. For each sample in a mini-batch, we select its K -nearest neighbours according to matrix D . We calculate the similarity between samples x_i and x_j ($i \neq j$) as follows:

$$D_{ij} = \frac{z_i^T \cdot z_j^T}{\|z_i^T\|_2 \cdot \|z_j^T\|_2} > 0, \quad (4)$$

where $\langle \cdot, \cdot \rangle$ is the dot product operation and $\|\cdot\|_2$ is the L_2 normalization. Eq. (4) computes the cosine value of

two samples; the larger the cosine value, the more similar the two samples are. We select the samples that correspond to the largest K values in matrix D as the K -nearest neighbours.

2.3. Neighbourhood Feature Relationship Distillation

We align the neighbourhood relationship based on the selected neighbours. We do not rely on a single value to measure the similarity between two samples. The feature maps are generated by convolving the original image, and they also serve as the input of subsequent layers. We exploit the rich similarity information from the feature maps. For the feature maps of sample x_i and its neighbour x_j , we construct the spatial similarity as follows:

$$\varphi_{ij} = \frac{\phi(|f_i - f_j|^2)}{\|\phi(|f_i - f_j|^2)\|_2}, \quad (5)$$

where $f_i \in R^{C \times H \times W}$ represents feature maps of x_i , C is the number of channels, H and W are the height and weight of feature maps, and $\phi(\cdot)$ represents the channel-wise average pooling. $\varphi_{ij} \in R^{H \times W}$ is the element-wise similarity of sample x_i and x_j . Eq. (5) is inspired by [14]. Therefore, φ_{ij} is essentially a similarity attention map. For K -nearest neighbours of sample x_i , we can construct the similarity for the layer l of teacher and student, $\varphi_i^{T(l)} = \{\varphi_{i1}^{T(l)}, \dots, \varphi_{iK}^{T(l)}\}$ and $\varphi_i^{S(l)} = \{\varphi_{i1}^{S(l)}, \dots, \varphi_{iK}^{S(l)}\}$. If the spatial resolution between teacher and student feature maps is different, we use adaptive pooling to align the resolution.

The neighbourhood feature relationship distillation loss can be calculated as Eq. (6):

$$L_{NFRD} = \frac{1}{N \times K} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^K \|\varphi_{ij}^{T(l)} - \varphi_{ij}^{S(l)}\|_2^2, \quad (6)$$

where K is the number of neighbours, and L is the number of layers.

2.4. Neighbourhood Logits Relationship Distillation

For the final output logits, we measure the similarity between the sample x_i and its neighbour x_j as follows. We use softmax function to transform the similarity into a probability distribution [15]:

$$\rho_{ij} = \text{softmax}(z_i - z_j), \quad (7)$$

where $\rho_{ij} \in R^{1 \times M}$, and M is the number of classes. We denote the logits similarity as a probability distribution. Each value in ρ_{ij} reflects the similarity of the network's assessment of sample x_i and its neighbour x_j for a certain category. Also, for K -nearest neighbours of sample x_i , we construct the similarity for the teacher and the student network, $\rho_i^T = \{\rho_{i1}^T, \dots, \rho_{iK}^T\}$ and $\rho_i^S = \{\rho_{i1}^S, \dots, \rho_{iK}^S\}$.

Our goal is not to make the network have the same assessment about samples x_i and x_j , but to maintain consistency between the two samples in the category similarity of the teacher and student networks. Therefore, we construct the following loss:

$$L_{NLRD} = \frac{1}{N \times K} \sum_{i=1}^N \sum_{j=1}^K JS(\rho_{ij}^T, \rho_{ij}^S), \quad (8)$$

where $JS(\cdot, \cdot)$ represents Jensen-Shannon divergence. Our method is applied at different layers to minimise the difference of the neighbourhood relationship between the teacher and student networks. The overall loss of training the student network can be formulated as follows:

$$L_{total} = L_{CE} + L_{KL} + \alpha L_{NFRD} + \beta L_{NLRD}, \quad (9)$$

where α and β indicate the loss weights. L_{CE} is the Cross-Entropy loss between predictions and ground truth labels.

3. EXPERIMENTS

3.1. Datasets and Setups

CIFAR100 [12] is a dataset that consists of 100 categories, with 500 training images and 100 testing images per category. For experiments on CIFAR100, we train all the models with SGD optimizer for 240 epochs. We use the batch size of 64. The learning rate starts from 0.05 except 0.01 for ShuffleNetV1 [16] and reduces to its 1/10 at 150, 180, 210 epochs. The weight decay and momentum are set to $5e-4$ and 0.9, respectively. We set the temperature to 4.

Tiny ImageNet [13] consists of 200 categories, which is a subset of ImageNet [17]. Each category has 500 training images, 50 validation images. We adopt SGD optimizer to train all the models for 200 epochs with batch size 64. The learning rate is initialized as 0.05 and divided by 10 at 60, 120, and 160 epochs. We set the weight decay and momentum as $5e-4$ and 0.9. The temperature is 4.

3.2. Comparisons with Recent Methods

We compare the proposed method with SOTA distillation methods. In KD [2], where the student network is supervised by soft targets of the teacher network. AT [14] transfers knowledge from attention maps. MGD [6] forces the student network to generate the teacher network's full feature with masked feature. RKD [7] penalizes the structure differences by distance and angle. ICKD [18] transfers channel relationship knowledge. Our method transfers neighbourhood relationship. For our method, we set the K (number of neighbours) to 1. We carried out experiments on CIFAR100 [12] and Tiny ImageNet [13]. We use models from many architectures, such as ResNet [19, 20], ShuffleNetV1 [16], and VGG [21].

Table 1. Top-1 accuracy (%) results on CIFAR100 test dataset.

Method	ResNet32×4 ShuffleNetV1	VGG13 VGG8	ResNet50 VGG8	ResNet32×4 ResNet8×4
Teacher	79.42	74.64	79.34	79.42
Student	70.50	70.36	70.36	72.50
KD [2]	74.96	73.02	73.33	73.68
AT [14]	73.06	71.91	72.01	74.09
RKD [7]	73.30	71.55	71.72	72.33
ICKD [18]	74.87	74.07	73.68	74.92
MGD [6]	N/A	73.37	74.08	75.54
Ours	76.70	74.62	75.05	76.28

Results on CIFAR100. As shown in Table 1, our method achieves superior performance over other knowledge distillation methods. For example, our method achieves a 76.70% classification accuracy for the (ResNet32×4, ShuffleNetV1) teacher-student pair, which is 1.74% and 1.83% higher than KD [2] and ICKD [18], respectively. Furthermore, we visualize the feature representations of the penultimate layer (before logits) using t-SNE [22]. In Fig. 2, the different colours indicate different categories on CIFAR100 dataset. Our method exhibits clearer separation between classes, while samples from the same class are closer to each other, in comparison with MGD [6]. This implies that our method enables the student to learn more discriminative feature representations.

Results on Tiny ImageNet. Table 2 shows the results of the Tiny ImageNet validation set. We report the Top-1 accuracy (%). From Table 3 we can also see that the proposed method achieves better performance than other distillation methods. For the (ResNet34, ResNet18) pair, our method achieves 68.93%, which improves the performance of KD [2] by 1.48%, and 0.43% higher than MGD [6], 1.15% higher than ICKD [18].

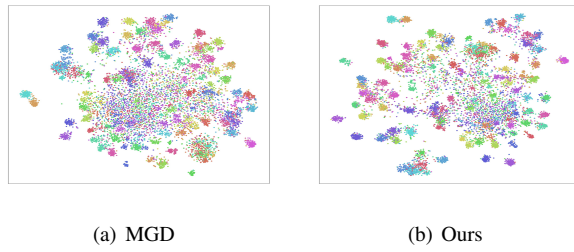


Fig. 2. Visualisation of the feature representations from the student network on CIFAR100. We use the student VGG8 and the teacher is ResNet50.

Table 2. Top-1 accuracy (%) results on Tiny ImageNet validation set.

Method	ResNet34 ResNet18	ResNet50 ResNet18
Teacher	67.71	68.88
Student	65.95	65.95
KD [2]	67.45	67.92
AT [14]	66.75	67.31
RKD [7]	66.59	67.74
ICKD [18]	67.78	69.18
MGD [6]	68.50	69.03
Ours	68.93	69.42

3.3. Ablation Studies

Loss Function. We verify the effectiveness of each part in the proposed method on CIFAR100 [12]. We record the results on four loss cases. As we can see from Table 3 that both L_{NFRD} and L_{NLRD} improve the performance compared with the baseline. We can see from the (ResNet32×4, ResNet8×4) teacher-student pair, L_{NFRD} has a 1.42% gain, L_{NLRD} has a 1.49% gain. When the two losses are combined, the performance is significantly improved. This demonstrates that both losses are effective.

Table 3. Effectiveness of each part. We report the Top-1 accuracy (%) on CIFAR100.

Teacher	Student	L_{NFRD}	L_{NLRD}	Top-1
ResNet32×4	ResNet8×4	×	×	73.68
		✓	×	75.10
		×	✓	75.17
		✓	✓	76.28

K -Nearest Neighbours. We also study the influence of the number of neighbours on our method. We conduct this experiment on CIFAR100 [12] and Tiny ImageNet [13].

As shown in Fig. 3, the accuracy of Tiny ImageNet is not sensitive to the number of neighbours, with a slight upward

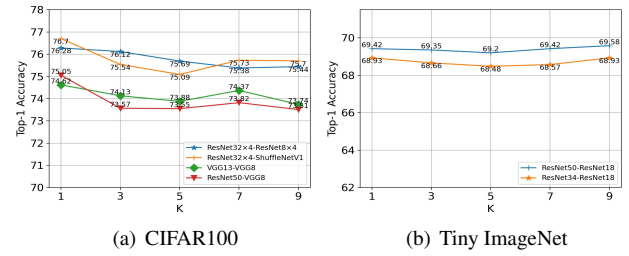


Fig. 3. The Top-1 accuracy of our method for various numbers of neighbours.

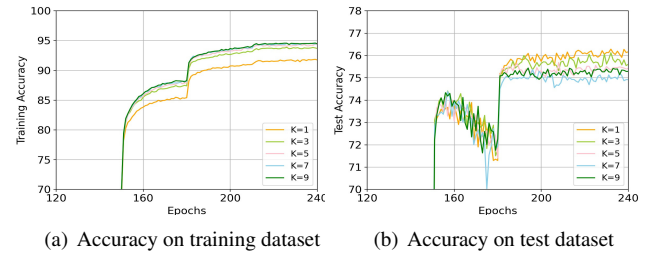


Fig. 4. The Top-1 accuracy curves of ResNet8×4 on CIFAR100.

trend as K increases. On CIFAR100, the optimal number of neighbours is 1, which yields the highest accuracy. Fig. 4 illustrates the accuracy curves of ResNet8×4 on CIFAR100 with different numbers of neighbours. We observe that more neighbours tend to result in better training accuracy. In particular, the training accuracy at $K = 9$ is much higher than that at $K = 1$. However, the test accuracy does not benefit from more nearest neighbours. As this shows, increasing the number of neighbours could lead to overfitting. Moreover, a lower number of neighbours can save storage space and computation time. Hence, our method attains the optimal benefit with the number of neighbours as 1.

4. CONCLUSION

In this paper, we propose a new similarity-based relational knowledge distillation method, which explores the knowledge of neighbourhood relationship. We argue that the relationship of the neighbourhood is concise, easy to learn, and transfers rich similarity information to the student. The proposed method is newly characterized and well distilled from both spatial similarity information of intermediate feature maps and similarity distribution information of the final logits. Extensive experiments show that our approach achieves better performance compared with SOTA methods.

5. REFERENCES

- [1] Jianping Gou, Baosheng Yu, Stephen J Maybank, and Dacheng Tao, "Knowledge distillation: A survey," *International Journal of Computer Vision*, vol. 129, pp. 1789–1819, 2021.
- [2] Geoffrey Hinton, Oriol Vinyals, Jeff Dean, et al., "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.
- [3] Ru Peng, Yawen Zeng, and Jake Zhao, "Distill the image to nowhere: Inversion knowledge distillation for multimodal machine translation," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022, pp. 2379–2390.
- [4] Weisong Zhao, Xiangyu Zhu, Kaiwen Guo, Xiao-Yu Zhang, and Zhen Lei, "Grouped knowledge distillation for deep face recognition," *arXiv preprint arXiv:2304.04462*, 2023.
- [5] Ziyao Guo, Haonan Yan, Hui Li, and Xiaodong Lin, "Class attention transfer based knowledge distillation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 11868–11877.
- [6] Zhendong Yang, Zhe Li, Mingqi Shao, Dachuan Shi, Zehuan Yuan, and Chun Yuan, "Masked generative distillation," in *Computer Vision–ECCV 2022*. Springer, 2022, pp. 53–69.
- [7] Wonpyo Park, Dongju Kim, Yan Lu, and Minsu Cho, "Relational knowledge distillation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 3967–3976.
- [8] Frederick Tung and Greg Mori, "Similarity-preserving knowledge distillation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 1365–1374.
- [9] Fei Ding, Yin Yang, Hongxin Hu, Venkat Krovi, and Feng Luo, "Dual-level knowledge distillation via knowledge alignment and correlation," *IEEE Transactions on Neural Networks and Learning Systems*, pp. 1–11, 2022.
- [10] Jianping Gou, Liyuan Sun, Baosheng Yu, Shaohua Wan, Weihua Ou, and Zhang Yi, "Multilevel attention-based sample correlations for knowledge distillation," *IEEE Transactions on Industrial Informatics*, vol. 19, no. 5, pp. 7099–7109, 2023.
- [11] Seyed-Iman Mirzadeh, Mehrdad Farajtabar, Ang Li, Nir Levine, Akihiro Matsukawa, and Hassan Ghasemzadeh, "Improved knowledge distillation via teacher assistant: Bridging the gap between student and teacher," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020, vol. 34, pp. 5191–5198.
- [12] Alex Krizhevsky, Geoffrey Hinton, et al., "Learning multiple layers of features from tiny images," 2009.
- [13] Ya Le and Xuan Yang, "Tiny imagenet visual recognition challenge," *CS 231N*, vol. 7, no. 7, pp. 3, 2015.
- [14] Nikos Komodakis and Sergey Zagoruyko, "Paying more attention to attention: improving the performance of convolutional neural networks via attention transfer," in *International Conference on Learning Representations*, 2017.
- [15] Soroush Abbasi Koohpayegani, Ajinkya Tejankar, and Hamed Pirsiavash, "Compress: Self-supervised learning by compressing representations," in *Advances in Neural Information Processing Systems*, 2020, vol. 33.
- [16] Xiangyu Zhang, Xinyu Zhou, Mengxiao Lin, and Jian Sun, "Shufflenet: An extremely efficient convolutional neural network for mobile devices," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 6848–6856.
- [17] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2009, pp. 248–255.
- [18] Li Liu, Qingle Huang, Sihao Lin, Hongwei Xie, Bing Wang, Xiaojun Chang, and Xiaodan Liang, "Exploring inter-channel correlation for diversity-preserved knowledge distillation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 8251–8260.
- [19] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [20] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun, "Identity mappings in deep residual networks," in *Computer Vision–ECCV 2016*. Springer, 2016, pp. 630–645.
- [21] Karen Simonyan and Andrew Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556*, 2014.
- [22] Laurens Van der Maaten and Geoffrey Hinton, "Visualizing data using t-sne," *Journal of machine learning research*, vol. 9, no. 86, pp. 2579–2605, 2008.